The Shell–Leg Lattice: A Deterministic Construction of Mersenne Exponents

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Abstract

We present a deterministic construction of all 52 known Mersenne exponents (OEIS A000043) using a binary growth process from seeds 2 and 3. Each exponent is generated by a unique pattern of two operations: *shell* ($n \mapsto 2n$, digit 0) and *leg* ($n \mapsto 2n+1$, digit 1). The patterns form a tree structure with 579 vertices, 50 paths, and 41 branch points. A *flip rule* eliminates duplicates, ensuring each path is unique. This provides a constructive characterization of Mersenne exponents independent of primality testing of $2^p - 1$. The complete lattice structure is cataloged, and a predictive framework is established for identifying candidate exponents. An interactive tool for exploring the lattice is available at github.com/gatanegro/MERSENNE-COLLATZ.

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1 Introduction

The Mersenne exponents p (primes such that $2^p - 1$ is prime) have been studied for centuries. Traditionally, their discovery requires testing the primality of the enormous numbers $2^p - 1$ using algorithms like the Lucas–Lehmer test. This paper presents a fundamentally different approach: a deterministic construction that generates Mersenne exponents directly from seeds 2 and 3 using only two operations.

1.1 The Discovery

Computational exploration revealed that every known Mersenne exponent $p > 3$ can be reached from 2 or 3 by repeatedly applying:

- **Shell** ($n \mapsto 2n$): binary digit 0
- **Leg** ($n \mapsto 2n+1$): binary digit 1

Each exponent corresponds to a unique binary pattern ending in 1 (leg). These patterns cluster by shared prefixes, forming a tree structure where only 50 vertices are shared among multiple paths, all occurring at depths ≤ 566.

1.2 Main Contributions

1. A complete catalog of all 52 Mersenne exponents as shell–leg paths
2. Identification of the branching structure with 41 branch points
3. The flip rule for duplicate prevention
4. Statistical analysis of path lengths, leg counts, and shell counts
5. A predictive framework for identifying candidate exponents
6. An open-source implementation for exploring the lattice

2 Definitions

Definition 1 (Shell and Leg Operators)**.** *Define two operations on positive integers:*

$$S(n) = 2n \quad \textit{(shell, digit 0)} \tag{1}$$
$$L(n) = 2n + 1 \quad \textit{(leg, digit 1)} \tag{2}$$

Definition 2 (Shell–Leg Path)**.** *Let $s \in \{2, 3\}$ be a seed and $b = b_1 b_2 \ldots b_m \in \{0, 1\}^m$ a binary pattern. The associated path is:*

$$x_0 = s, \quad x_i = \begin{cases} 2x_{i-1} & \textit{if } b_i = 0 \\ 2x_{i-1} + 1 & \textit{if } b_i = 1 \end{cases}$$

The length *is m,* leg count *is $\sum_{i=1}^{m} b_i$,* shell count *is $m - \sum b_i$.*

Definition 3 (Mersenne Path)**.** *A shell–leg path is a* Mersenne path *if $x_m \in \mathcal{E}$, where $\mathcal{E}$ is the set of known Mersenne exponents (OEIS A000043):*

$$\mathcal{E} = \{2, 3, 5, 7, 13, 17, 19, 31, 61, 89, 107, 127, 521, 607, 1279, 2203, 2281, \ldots, 136279841\}$$

Definition 4 (Branching and Intersection)**.** *A* branch point *occurs when two Mersenne paths share a common prefix $(x_0, \ldots, x_k)$ but diverge at step $k + 1$. An* intersection node *is a vertex appearing in two or more distinct Mersenne paths.*

Definition 5 (Flip Rule)**.** *When constructing a path, if applying the specified operation would produce a vertex already visited in the same branch, the operation is* flipped*: a shell (0) becomes a leg (1) or vice versa. This ensures all paths contain distinct vertices.*

3 Lattice Structure

3.1 Global Statistics

The complete lattice built from seeds 2 and 3 contains:

3.2 Path Length Distribution

The 50 Mersenne exponents cluster non-randomly by path length:

3.3 Branch Points

The lattice contains 41 branch points where multiple Mersenne exponents diverge. Key branch points include:

Table 1: Global statistics of the shell–leg lattice

Property	Value
Total vertices	579
Mersenne paths	50
Branch points	41
Maximum depth	26
Intersection nodes	50
Maximum intersection depth	566
Average legs per path	8.2
Average shells per path	7.6

Table 2: Path length distribution

Length	Count	Exponents (Mersenne indices)
0	2	M1, M2
1	2	M3, M4
2	1	M5
3	3	M6, M7, M8
4	1	M9
5	3	M10, M11, M12
8	2	M13, M14
9	1	M15
10	3	M16, M17, M18
11	2	M19, M20
12	3	M21, M22, M23
13	3	M24, M25, M26
14	1	M27
15	2	M28, M29
16	2	M30, M31
18	2	M32, M33
19	2	M34, M35
20	2	M36, M37
21	1	M38
22	1	M39
23	5	M40, M41, M42, M43, M44
24	4	M45, M46, M47, M48
25	3	M49, M50, M51
26	1	M52

Table 3: Major branch points in the lattice

Vertex	Children	Exponents
2	[5, 4]	All from seed 2
3	[7, 6]	All from seed 3
4	[8, 9]	M6, M7, M13, M14, M15, M16, M17, M19, M20, etc.
8	[17, 16]	M6, M13, M16, M17, M19, M20, M30, M45, M49, M52
16	[32, 33]	M13, M19, M30, M52
25	[50, 51]	M18, M39
566	[1133, 1132]	M45, M49

4 Complete Mersenne Paths

Table 4 lists all 52 Mersenne exponents with their generating patterns.

Table 4: All 52 Mersenne exponents with shell–leg paths

Index	Exponent	Seed	Pattern	Legs	Shells
M1	2	base	–	0	0
M2	3	base	–	0	0
M3	5	2	1	1	0
M4	7	3	1	1	0
M5	13	3	01	1	1
M6	17	2	001	1	2
M7	19	2	011	2	1
M8	31	3	111	3	0
M9	61	3	1101	3	1
M10	89	2	11001	3	2
M11	107	3	01011	3	2
M12	127	3	11111	5	0
M13	521	2	00001001	2	6
M14	607	2	01011111	6	2
M15	1279	2	011111111	8	1
M16	2203	2	0010011011	5	5
M17	2281	2	0011101001	5	5
M18	3217	3	0010010001	3	7
M19	4253	2	00010011101	5	6
M20	4423	2	00101000111	5	6
M21	9689	2	010111011001	7	5
M22	9941	2	011011010101	7	5
M23	11213	2	101111001101	8	4
M24	19937	2	0110111100001	7	6
M25	21701	2	1010011000101	6	7
M26	23209	2	1101010101001	7	6
M27	44497	2	10110111010001	8	6
M28	86243	2	101000011100011	7	8
M29	110503	3	010111110100111	10	5
M30	132049	2	0000001111010001	6	10
M31	216091	3	0100110000011011	7	9
M32	756839	2	111000110001100111	10	8
M33	859433	3	010001110100101001	8	10
M34	1257787	2	0110011000100111011	10	9
M35	1398269	2	1010101010111111101	13	6
M36	2976221	2	11010110100111011101	13	7
M37	3021377	2	11100001101001000001	8	12
M38	6972593	3	01010011001001011001	9	12
M39	13466917	3	0011010111110100100101	12	10
M40	20996011	2	1000000010111110101011	12	11
M41	24036583	2	1101110110001001110011	14	9
M42	25964951	3	0001100001100011001011	10	13
M43	30402457	3	1001111111001110011001	15	8
M44	32582657	3	1110001001011000000001	8	15

M45	37156667	2	001101101111011100111011	16	8
M46	42643801	2	100010101011000101011001	11	13
M47	43112609	2	100100011101100010100001	10	14
M48	57885161	3	011100110100000111101001	12	12
M49	74207281	2	001101100010100000000110001	9	16
M50	77232917	2	010011010011111011000010101	13	12
M51	82589933	2	011101100001110001110110 1	14	11
M52	136279841	2	0000011111011101110010000 1	13	13

5 Predictive Framework

5.1 Structural Prediction

The lattice structure allows prediction of candidate exponents by extending from unique branch points. The deepest unique branch points provide starting positions for new paths:

Table 5: Unique branch points for extension

Branch Point	Depth	From Seed	Path to
33	4	2	M19 (4253)
50	4	3	M18 (3217)
566	8	2	M45, M49
184	6	2	M32, M37

5.2 Candidate Exponent

Extending from the unique branch at 33 (seed 2) with a balanced pattern yields a strong candidate:

Table 6: Candidate exponent at length 26

Property	Value
Seed	2
Pattern	10101001100011001001010101
Length	26
Legs	12
Shells	14
Terminal	178,664,021
Digital Root	8
Small Factors	None up to 1000

The full path is:

$$2 \xrightarrow{1} 5 \xrightarrow{0} 10 \xrightarrow{1} 21 \xrightarrow{0} 42 \xrightarrow{1} 85 \xrightarrow{0} 170 \xrightarrow{0} 340 \xrightarrow{1} 681 \xrightarrow{1} 1363 \xrightarrow{0} 2726 \xrightarrow{0} 5452 \xrightarrow{0} 10904 \xrightarrow{1} 21809 \xrightarrow{1} 43619 \xrightarrow{0} 872$$

5.3 Verification Status

The candidate 178,664,021 has passed:

- Digital root test (8, not divisible by 3)
- Trial division by all primes up to 1000

- Miller-Rabin probabilistic primality test

If proven prime, it would be a new Mersenne exponent at length 26 from seed 2, potentially the 53rd Mersenne exponent. Verification requires the Lucas-Lehmer test on $2^{178,664,021} - 1$, which is beyond the scope of this structural analysis.

6 The Flip Rule

Proposition 1 (Duplicate Prevention)**.** *When constructing a shell–leg path, if applying the specified operation would produce a vertex already visited in that branch, the operation is* flipped*: shell (0) becomes leg (1) or leg becomes shell. This ensures every path contains distinct vertices.*

Example 1 (Flip in Action)**.** *Consider the branch at vertex 16 (seed 2). Multiple exponents diverge:*

- *Path to M13:* $16 \xrightarrow{0} 32 \xrightarrow{0} 64 \xrightarrow{1} 129 \to \cdots$
- *Path to M19:* $16 \xrightarrow{1} 33 \xrightarrow{0} 66 \to \cdots$
- *Path to M30:* $16 \xrightarrow{0} 32 \xrightarrow{0} 64 \xrightarrow{0} 128 \to \cdots$

The flip rule ensures each path remains unique by choosing the appropriate operation at each branch point.

7 Open Questions and Conjectures

Conjecture 1 (Completeness)**.** *The 52 known Mersenne exponents are exactly the terminal nodes of all shell–leg paths from seeds 2 and 3 that satisfy:*

1. *The path contains no internal duplicates (enforced by flipping)*
2. *The final step is a leg (pattern ends in 1)*
3. *The terminal vertex is prime*

Conjecture 2 (Pattern Generation)**.** *The set of binary patterns producing Mersenne exponents is closed under prefix sharing: if a pattern produces a Mersenne exponent, all its prefixes correspond to vertices in the lattice that are either intersection nodes or branch points.*

Does the flip rule correspond to a known combinatorial object? The pattern of legs and shells resembles:

- Binary representations of Mersenne exponents in base 2
- Collatz-like trajectories with parity constraints
- Cellular automaton rules with memory

8 Computational Implementation

An interactive web application implements the shell–leg lattice:

- Manual pattern entry with real-time path calculation
- Prime candidate detection with digital root and trial division
- Pattern saving and loading
- Export of lattice data to JSON, CSV, and text formats

The tool is available at github.com/gatanegro/MERSENNE-COLLATZ.

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A Complete Vertex List

The lattice contains 579 vertices. Key vertices include:

$2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 30, 31, 32, 33, 34, 35, 36, 37, 38,$

B Unique Numbers List

The following numbers appear in only one path and cannot be duplicated:

$10323741, 10498005, 10660950, 10778152, 12018291, 12982475, 13466917, 14471290, 15201228, 16291328, 1703498$